

L3.1 Convolutional Neural Networks Quiz

ANS

1. What does a convolution filter produce as it slides across an input volume?

- A) A two-dimensional feature map
- B) A class label directly
- C) A smaller training dataset
- D) A fully connected layer

ANS: A) A two-dimensional feature map

2. How is one convolution output value computed?

- A) By selecting the smallest input value
- B) By elementwise multiplication followed by summation
- C) By sorting all image pixels
- D) By averaging all feature maps

ANS: B) By elementwise multiplication followed by summation

3. In a CNN, what typically performs feature extraction before classification?

- A) Only fully connected layers
- B) Only the SoftMax layer
- C) CONV, pooling, and nonlinear activation layers
- D) The loss function alone

ANS: C) CONV, pooling, and nonlinear activation layers

4. For a convolution on an RGB input, what must equal the filter depth?

- A) The filter width
- B) The output width
- C) The number of filters
- D) The number of input channels

ANS: D) The number of input channels

5. If a convolution layer uses six filters, how many output feature-map channels does it produce?

- A) Six
- B) One
- C) Three
- D) The same number as the filter width

ANS: A) Six

6. What output spatial size results from a 7 x 7 input, a 3 x 3 filter, stride 1, and no padding?

- A) 7 x 7
- B) 5 x 5
- C) 4 x 4
- D) 3 x 3

ANS: B) 5 x 5

7. What is a primary purpose of padding in a convolution layer?

- A) To remove all image boundaries
- B) To increase the number of input channels
- C) To control output size and allow border pixels to be processed
- D) To replace the activation function

ANS: C) To control output size and allow border pixels to be processed

8. Which setting preserves spatial width and height for a 3 x 3 convolution with stride 1?

- A) Padding 0
- B) Padding 2
- C) Stride 3
- D) Padding 1

ANS: D) Padding 1

9. What does a 1 x 1 pointwise convolution primarily operate across at each spatial location?

- A) The channel dimension
- B) Only the image width
- C) Only the image height
- D) The training-batch dimension

ANS: A) The channel dimension

10. What is the main benefit of dilated convolution described in the slides?

- A) It removes the need for filters
- B) It increases the receptive field without increasing the parameter count
- C) It always reduces the number of channels to one
- D) It converts an image into a class label

ANS: B) It increases the receptive field without increasing the parameter count

11. For which data is 3D convolution especially useful?

- A) One-dimensional text only
- B) Scalar sensor readings only
- C) Medical CT/MRI volumes and lidar point clouds
- D) Tabular spreadsheets only

ANS: C) Medical CT/MRI volumes and lidar point clouds

12. Why can converting convolution to matrix multiplication improve computational efficiency?

- A) It eliminates the convolution weights
- B) It requires no memory
- C) It avoids all parallel computation
- D) GPUs and FPGAs can process matrix multiplication efficiently

ANS: D) GPUs and FPGAs can process matrix multiplication efficiently

13. What operation does max pooling perform on each local region?

- A) It selects the largest value
- B) It selects the smallest value
- C) It computes a class probability
- D) It creates trainable filter weights

ANS: A) It selects the largest value

14. Given the 2 x 2 region [1, 1; 5, 6], what is its max-pooling output?

- A) 1
- B) 6
- C) 3.25
- D) 13

ANS: B) 6

15. How does a CNN generally differ from an MLP in parameter usage?

- A) A CNN always has more parameters
- B) An MLP uses convolution in every layer
- C) A CNN typically uses fewer parameters through local connectivity and sharing
- D) They always have exactly the same number of parameters

ANS: C) A CNN typically uses fewer parameters through local connectivity and sharing

16. Which architecture is identified as the best-performing VGGNet variant in the slides?

- A) VGG-5
- B) VGG-8
- C) VGG-19 only
- D) VGG-16

ANS: D) VGG-16

17. How many object classes are included in the ImageNet Large Scale Visual Recognition Challenge described in the slides?

- A) 1,000
- B) 100
- C) 10,000
- D) 1.4 million

ANS: A) 1,000

18. Compared with AlexNet, why does GoogLeNet have far fewer parameters according to the slides?

- A) It uses no convolution layers
- B) It has no fully connected layers
- C) It processes only grayscale images
- D) It uses only one filter

ANS: B) It has no fully connected layers

19. What sequence defines depthwise separable convolution?

- A) Pooling followed by SoftMax
- B) A fully connected layer followed by pooling
- C) Depthwise convolution followed by pointwise convolution
- D) Two fully connected layers with no convolution

ANS: C) Depthwise convolution followed by pointwise convolution

20. What is normally the first modification when applying transfer learning to a pretrained CNN for a new classification task?

- A) Delete all convolution layers
- B) Replace every filter with a handcrafted edge detector
- C) Increase the input to an arbitrary number of channels
- D) Replace the SoftMax classification head with one for the new task

ANS: D) Replace the SoftMax classification head with one for the new task